

TASK-2

Migrating data from RDS to HBASE using SQOOP

Step1. Switching to hbase shell and creating table.

```
hbase shell
create 'hbtaxi','td'
```

```
[hbase(main):001:0> create 'hbtaxi','td'
0 row(s) in 2.5200 seconds

=> Hbase::Table - hbtaxi
```

Step 2. Switching back to cluster and installing mysql connector on the cluster

```
Wget https://de-mysql-connector.s3.amazonaws.com/mysql-connector-java-8.0.25.tar.gz
tar -xvf mysql-connector-java-8.0.25.tar.gz
cd mysql-connector-java-8.0.25/
sudo cp mysql-connector-java-8.0.25.jar /usr/lib/sqoop/lib/
```

```
[hadoop@ip-172-30-0-108 ~]$ wget https://de-mysql-connector.s3.amazonaws.com/mysql-connector-java-8.0.25.tar.gz
--2023-10-07 10:08:29-- https://de-mysql-connector.s3.amazonaws.com/mysql-connector-java-8.0.25.tar.gz
Resolving de-mysql-connector.s3.amazonaws.com (de-mysql-connector.s3.amazonaws.com)... 52.217.41.188, 52.217.117.65, 54.231.129.193, ...
Connecting to de-mysql-connector.s3.amazonaws.com (de-mysql-connector.s3.amazonaws.com)|52.217.41.188|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4079310 (3.9M) [application/x-gzip]
Saving to: 'mysql-connector-java-8.0.25.tar.gz'
[
100%[=====] 4,079,310 --.-K/s in 0.1s
[
2023-10-07 10:08:29 (33.0 MB/s) - 'mysql-connector-java-8.0.25.tar.gz' saved [4079310/4079310]

[hadoop@ip-172-30-0-108 ~]$ tar -xvf mysql-connector-java-8.0.25.tar.gz
mysql-connector-java-8.0.25/
mysql-connector-java-8.0.25/src/
```

Step 3. Running the scoop command to import table. From RDS to hbase

```
scoop import --connect jdbc:mysql://database-1.cllbwj12j2jd.us-east-1.rds.amazonaws.com:3306/yellow --table taxi --hbase-table htaxi --column-family td --hbase-row-key VendorID,tpep_pickup_datetime,tpep_dropoff_datetime --split-by VendorID --username admin --password pass1234 --num-mappers 5 --hbase-bulkload;
```

```
[hadoop@ip-172-30-0-108 ~]$ scoop import --connect jdbc:mysql://database-1.cllbwj12j2jd.us-east-1.rds.amazonaws.com:3306/yellow --table taxi --hbase-table htaxi --column-family td --hbase-row-key VendorID,tpep_pickup_datetime,tpep_dropoff_datetime --split-by VendorID --username admin --password pass1234 --num-mappers 5 --hbase-bulkload;
Warning: /usr/lib/scoop/.accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
23/10/07 18:59:25 INFO scoop.Scoop: Running Scoop version: 1.4.7
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/share/aws/redshift/jdbc/redshift-jdbc42-1.2.37.1061.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/lib/hive/lib/log4j-slf4j-impl-2.6.2.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
23/10/07 18:59:25 WARN tool.BaseScoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
23/10/07 18:59:25 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
23/10/07 18:59:25 INFO tool.CodeGenTool: Beginning code generation
Loading class 'com.mysql.jdbc.Driver'. This is deprecated. The new driver class is 'com.mysql.cj.jdbc.Driver'. The driver is automatically registered via the SPI and manual loading of the driver class is generally unnecessary.
23/10/07 18:59:25 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM 'taxi' AS t LIMIT 1
23/10/07 18:59:25 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM 'taxi' AS t LIMIT 1
23/10/07 18:59:25 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /usr/lib/hadoop-mapreduce
Note: /tmp/scoop-hadoop/compile/41a0f5feb13dc5636fba736b016973e6/taxi.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
23/10/07 18:59:28 INFO orm.CompilationManager: Writing jar file: /tmp/scoop-hadoop/compile/41a0f5feb13dc5636fba736b016973e6/taxi.jar
23/10/07 18:59:28 WARN manager.MySQLManager: It looks like you are importing from mysql.
23/10/07 18:59:28 WARN manager.MySQLManager: This transfer can be faster! Use the --direct
23/10/07 18:59:28 WARN manager.MySQLManager: option to exercise a MySQL-specific fast path.
23/10/07 18:59:28 INFO manager.MySQLManager: Setting zero DATETIME behavior to convertToNull (mysql)
23/10/07 18:59:28 INFO mapreduce.ImportJobBase: Beginning import of taxi
23/10/07 18:59:28 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
23/10/07 18:59:28 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
23/10/07 18:59:30 WARN mapreduce.TableMapReduceUtil: The addDependencyJars(Configuration, Class<?..>) method has been deprecated since it is easy to use incorrectly. Most users should rely on addDependencyJars(Job) instead. See HBASE-8386 for more details.
23/10/07 18:59:30 WARN mapreduce.TableMapReduceUtil: The addDependencyJars(Configuration, Class<?..>) method has been deprecated since it is easy to use incorrectly. Most users should rely on addDependencyJars(Job) instead. See HBASE-8386 for more details.
23/10/07 18:59:30 INFO zlib.ZlibFactory: Successfully loaded & initialized native-zlib library
23/10/07 18:59:30 INFO compress.CodecPool: Got brand-new compressor [.deflate]
23/10/07 18:59:30 INFO client.RMProxy: Connecting to ResourceManager at ip-172-30-0-108.ec2.internal/172.30.0.108:8032
23/10/07 18:59:32 INFO db.DBInputFormat: Using read committed transaction isolation
23/10/07 18:59:32 INFO db.DataDrivenDBInputFormat: BoundingValsQuery: SELECT MIN('VendorID'), MAX('VendorID') FROM 'taxi'
23/10/07 11:00:09 INFO db.IntegerSplitter: Split size: 0; Num splits: 5 from: 1 to: 2
23/10/07 11:00:10 INFO mapreduce.JobSubmitter: number of splits:2
23/10/07 11:00:10 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1696672913558_0002
23/10/07 11:00:10 INFO impl.YarnClientImpl: Submitted application application_1696672913558_0002
23/10/07 11:00:10 INFO mapreduce.Job: The url to track the job: http://ip-172-30-0-108.ec2.internal:20888/proxy/application_1696672913558_0002/
23/10/07 11:00:10 INFO mapreduce.Job: Running job: job_1696672913558_0002
23/10/07 11:00:16 INFO mapreduce.Job: Job job_1696672913558_0002 running in uber mode : false
23/10/07 11:00:16 INFO mapreduce.Job: map 0% reduce 0%
23/10/07 11:00:04 INFO mapreduce.Job: map 37% reduce 0%
23/10/07 11:00:10 INFO mapreduce.Job: map 47% reduce 0%
23/10/07 11:00:16 INFO mapreduce.Job: map 50% reduce 0%
23/10/07 11:00:22 INFO mapreduce.Job: map 53% reduce 0%
23/10/07 11:07:04 INFO mapreduce.Job: map 67% reduce 0%
23/10/07 11:07:10 INFO mapreduce.Job: map 78% reduce 0%
23/10/07 11:07:16 INFO mapreduce.Job: map 83% reduce 0%
23/10/07 11:07:41 INFO mapreduce.Job: map 67% reduce 0%
23/10/07 11:07:45 INFO mapreduce.Job: map 83% reduce 0%
23/10/07 11:09:27 INFO mapreduce.Job: map 100% reduce 0%
23/10/07 11:09:53 INFO mapreduce.Job: map 100% reduce 47%
23/10/07 11:10:11 INFO mapreduce.Job: map 100% reduce 67%
23/10/07 11:10:35 INFO mapreduce.Job: map 100% reduce 68%
23/10/07 11:11:05 INFO mapreduce.Job: map 100% reduce 69%
23/10/07 11:11:30 INFO mapreduce.Job: map 100% reduce 70%
23/10/07 11:12:00 INFO mapreduce.Job: map 100% reduce 71%
23/10/07 11:12:30 INFO mapreduce.Job: map 100% reduce 72%

23/10/07 11:25:01 INFO mapreduce.Job: map 100% reduce 100%
23/10/07 11:25:11 INFO mapreduce.Job: Job job_1696672913558_0002 completed successfully
23/10/07 11:25:12 INFO mapreduce.Job: Counters: 49
File System Counters
  FILE: Number of bytes read=12400011487
  FILE: Number of bytes written=17042705536
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=221
  HDFS: Number of bytes written=24020406740
  HDFS: Number of read operations=13
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=5
Job Counters
  Launched map tasks=2
  Launched reduce tasks=1
  Other local map tasks=2
  Total time spent by all maps in occupied slots (ms)=44683290
  Total time spent by all reduces in occupied slots (ms)=84847590
  Total time spent by all map tasks (ms)=992962
  Total time spent by all reduce tasks (ms)=942751
  Total vcore-milliseconds taken by all map tasks=992962
  Total vcore-milliseconds taken by all reduce tasks=942751
  Total megabyte-milliseconds taken by all map tasks=1429865280
  Total megabyte-milliseconds taken by all reduce tasks=2715122800
Map-Reduce Framework
  Map input records=18880595
  Map output records=283208925
  Map output bytes=41823258041
  Map output materialized bytes=4642810134
  Input split bytes=221
  Combine input records=0
  Combine output records=0
  Reduce input groups=18856607
  Reduce shuffle bytes=4642010134
  Reduce input records=283208925
  Reduce output records=282850305
  Spilled Records=1039554478
  Shuffled Maps=2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=9876
  CPU time spent (ms)=2074300
  Physical memory (bytes) snapshot=2690126824
  Virtual memory (bytes) snapshot=119067136
  Total committed heap usage (bytes)=1080926432
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=24020406740
23/10/07 11:25:12 INFO mapreduce.ImportJobBase: Transferred 22.3707 GB in 1.541.4599 seconds (14.861 MB/sec)
23/10/07 11:25:12 INFO mapreduce.ImportJobBase: Retrieved 283208925 records.
23/10/07 11:25:12 WARN mapreduce.LoadIncrementalHFiles: managed connection cannot be used for bulkload. Creating unmanaged connection.
23/10/07 11:25:12 WARN mapreduce.LoadIncrementalHFiles: Skipping non-directory hdfs://ip-172-30-0-108.ec2.internal:8020/user/hadoop/taxi/_SUCCESS
23/10/07 11:25:12 INFO impl.MetricsConfig: loaded properties from hadoop-metrics2-hbase.properties
```

Step 4. Switching to Hbase shell and counting the records.

Count 'hbtaxi'

```
at org.jruby.Main.main(Main.java:188)
log4j:ERROR Either File or DatePattern options are not set for appender [DRFAS].
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
Version 1.4.13, rUnknown, Fri Apr 17 15:18:24 UTC 2020
```

```
[hbase(main):001:0> count 'hbtaxi'
Current count: 1000, row: 1_2017-01-01 00:11:36.0_2017-01-01 00:24:48.0
Current count: 2000, row: 1_2017-01-01 00:18:15.0_2017-01-01 00:26:33.0
Current count: 3000, row: 1_2017-01-01 00:23:58.0_2017-01-01 00:42:06.0
Current count: 4000, row: 1_2017-01-01 00:29:40.0_2017-01-01 00:46:41.0
Current count: 5000, row: 1_2017-01-01 00:35:28.0_2017-01-01 00:48:58.0
Current count: 6000, row: 1_2017-01-01 00:41:06.0_2017-01-01 01:00:39.0
Current count: 7000, row: 1_2017-01-01 00:46:43.0_2017-01-01 01:06:23.0
Current count: 8000, row: 1_2017-01-01 00:52:09.0_2017-01-01 00:59:09.0
Current count: 9000, row: 1_2017-01-01 00:57:37.0_2017-01-01 01:11:55.0
Current count: 10000, row: 1_2017-01-01 01:02:58.0_2017-01-01 01:09:05.0
Current count: 11000, row: 1_2017-01-01 01:08:31.0_2017-01-01 01:14:57.0
Current count: 12000, row: 1_2017-01-01 01:13:50.0_2017-01-01 01:25:21.0
```

```
Current count: 18820000, row: 2_2017-02-28 20:55:42.0_2017-02-28 21:04:10.0
Current count: 18828000, row: 2_2017-02-28 21:00:53.0_2017-02-28 21:14:57.0
Current count: 18829000, row: 2_2017-02-28 21:06:00.0_2017-02-28 21:43:54.0
Current count: 18830000, row: 2_2017-02-28 21:11:07.0_2017-02-28 21:16:43.0
Current count: 18831000, row: 2_2017-02-28 21:16:34.0_2017-02-28 21:37:53.0
Current count: 18832000, row: 2_2017-02-28 21:22:02.0_2017-02-28 21:36:41.0
Current count: 18833000, row: 2_2017-02-28 21:27:37.0_2017-02-28 21:35:04.0
Current count: 18834000, row: 2_2017-02-28 21:32:58.0_2017-02-28 21:39:03.0
Current count: 18835000, row: 2_2017-02-28 21:38:13.0_2017-02-28 21:47:53.0
Current count: 18836000, row: 2_2017-02-28 21:43:40.0_2017-02-28 21:51:15.0
Current count: 18837000, row: 2_2017-02-28 21:48:57.0_2017-02-28 21:52:51.0
Current count: 18838000, row: 2_2017-02-28 21:54:10.0_2017-02-28 22:04:31.0
Current count: 18839000, row: 2_2017-02-28 21:59:05.0_2017-02-28 22:17:20.0
Current count: 18840000, row: 2_2017-02-28 22:04:11.0_2017-02-28 22:19:20.0
Current count: 18841000, row: 2_2017-02-28 22:09:25.0_2017-02-28 22:16:11.0
Current count: 18842000, row: 2_2017-02-28 22:14:24.0_2017-02-28 22:43:03.0
Current count: 18843000, row: 2_2017-02-28 22:19:55.0_2017-02-28 22:32:38.0
Current count: 18844000, row: 2_2017-02-28 22:25:33.0_2017-02-28 22:33:22.0
Current count: 18845000, row: 2_2017-02-28 22:31:29.0_2017-02-28 22:45:52.0
Current count: 18846000, row: 2_2017-02-28 22:37:51.0_2017-02-28 22:51:27.0
Current count: 18847000, row: 2_2017-02-28 22:44:04.0_2017-02-28 22:59:46.0
Current count: 18848000, row: 2_2017-02-28 22:50:35.0_2017-02-28 22:56:35.0
Current count: 18849000, row: 2_2017-02-28 22:57:20.0_2017-02-28 23:02:07.0
Current count: 18850000, row: 2_2017-02-28 23:04:48.0_2017-02-28 23:16:46.0
Current count: 18851000, row: 2_2017-02-28 23:12:03.0_2017-02-28 23:17:59.0
Current count: 18852000, row: 2_2017-02-28 23:19:06.0_2017-02-28 23:55:06.0
Current count: 18853000, row: 2_2017-02-28 23:26:26.0_2017-02-28 23:27:52.0
Current count: 18854000, row: 2_2017-02-28 23:34:40.0_2017-02-28 23:38:29.0
Current count: 18855000, row: 2_2017-02-28 23:43:24.0_2017-02-28 23:54:31.0
Current count: 18856000, row: 2_2017-02-28 23:53:00.0_2017-03-01 00:27:20.0
18856687 row(s) in 457.6550 seconds
```

```
=> 18856687
```

```
hbase(main):002:0> █
```

Hence, all records were imported from RDS to Hbase successfully.